

Section 2

Digital Transformation across Industries

Chapter 5

How Telecom Players can Win in the Digital Age

- Telcos have been investing a lot in digital infrastructure as the industry progressed from 2G to 5G. However, they have not realized the desired value from these investments.
- Instead, OTTs have won the game of content, leveraging the infrastructure that telcos had primarily invested in.
- Now, 5G presents telcos a breakthrough opportunity. But to realize gains from it, they will need to reshape their offerings, their business model and internal processes.

Key Takeaways

Key challenges for telecom industry

Why telecom operators are obsessed with deployment of 5G?

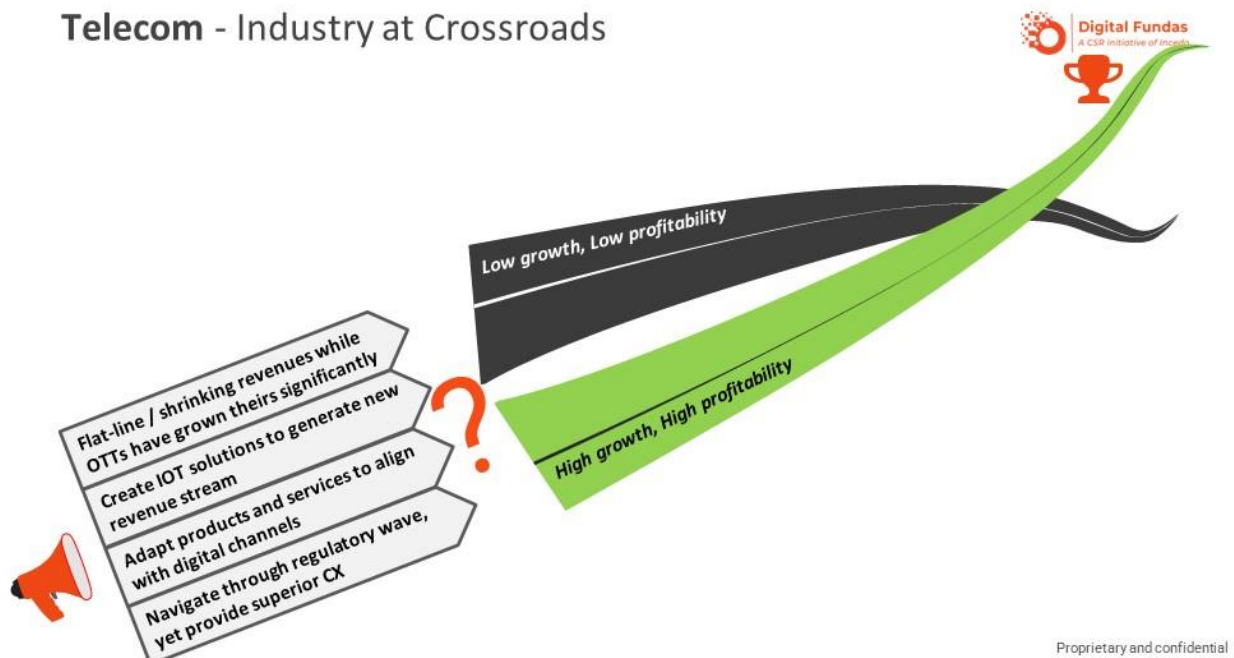
Lesson 1: Four key challenges for the Telecom Industry

Nitin's introduction:

The telecommunications Industry has evolved around technology advancements in the communications space. The industry has developed from being a provider of landline services to delivering high-speed broadband services with ubiquitous mobility.

The telco industry today is certainly at a point of inflexion, and in order to remain relevant in the rapidly changing marketplace, its long-standing aspiration of evolving from the legacy of providing connectivity services to a more value-added provider of content, applications and data services has to be accomplished. In fact, the rapid adoption of digital transformation by enterprises is a great opportunity for telcos globally. For telcos, it's not just about the digital transformation that they are themselves undergoing, it is also about their being the digital enablers for other enterprises and sectors. GSMA has projected that the number of connected

devices will grow from 7 billion in 2018 to 25 billion in 2025;¹³ and telecom operators are in a unique position to capitalize on this by providing innovative digital connectivity solutions that will form the backbone of the globally connected digital world in the near future.



1. **Despite growing data consumption, telcos have been losing the content and application battle to OTT (over-the-top) players, leading to flat-line/shrinking revenues.**

In general business terms, increasing customer demand leads to an increase in revenues, but for telecom operators, data usage has increased exponentially while revenues at best have remained flatlined. Capex investments made by telcos in 3G and 4G in the last decade have led to a major technological shift in the form of high-speed global mobile networks. Most of these investments have enabled entry to OTT players who offer consumers innovative services. While OTT players who fully leverage the connectivity infrastructure provided by telecom operators have grown their revenues exponentially over the past decade, the telecom operators haven't been able to monetize this phenomenon and bite into any incremental revenues for themselves.

2. **Massive investments are being made in 5G infrastructure; however, monetization through IOT solutions and the ability of telcos to create new revenue streams is unclear.**

When it comes to new revenues streams for telecom operators, IOT is high on the list. After all, IOT is the big phenomenon which is likely to create billions of connected devices across the world powered by these telecom operators. While there is no doubt about the potential of the 5G + IOT combination, the big question is, how are the telcos going to capitalize on this and generate revenues for themselves. Despite a number of trials and pilot implementations, it is not clear what role the telco can play or what business model it can have beyond that of a connectivity provider.

3. **Digital channels have not yielded the desired results; digital contributes less than 15 per cent of total sales and services for many leading telcos.**

Telcos have made very significant investments in digital transformation over the last five to seven years. However, these investments have not yielded the desired outcomes, and customers continue to rely heavily on stores and call centres for their needs. The main reason for this is that telcos have not adapted their products and services for digital channels. In this industry, policies, the fine print and pricing continue to be complex, necessitating the need for assisted sales. Moreover, the telcos' digital journeys have mirrored physical journeys, making them long and cumbersome. Another challenge the telcos face is that while they generate huge amounts of data on customer usage, the systems of insights to convert this data to create personalized experiences for the consumer have not evolved at the required pace.

4. **Keeping up with cybersecurity, regulatory and government policy changes**

The focus of regulators is increasingly turning towards areas such as general data protection regulation (GDPR), while legacy issues such as wholesale pricing and VoIP interconnection regulations are easing. But other areas of regulation are changing very fast, and we have spectrum release road maps, data protection frameworks and accounting rules, all in ever-evolving states.

Monitoring call and communication quality is crucial for ensuring that customers get the kind of telecom experience they deserve. Tracking and understanding their communication life cycle will be essential for telecom companies for another reason too – which is to do with data and privacy. Customers are terrified at the possibility of identity theft and financial loss. As a result, customers are demanding the same level of control over their personal data as they have over passwords to their bank accounts and email addresses. Telcos will need to work harder to create environments where people can feel safe.

Telco operators are working towards providing personalized experiences leveraging AI/ML and digital analytics capabilities, which, from a customer-centric point of view, is the right thing to do. However, customers don't want to compromise on how their data is protected and stored. Telcos will need to ensure that their networks can protect consumer data in line with industry standards and to the satisfaction of their customers.

Nitin's conclusion:

While there are headwinds, and given that the telecom industry is at a point of inflection, telcos have inherent strengths that they can use to turn the tables to their advantage. Telecom players have huge physical assets that act as an entry barrier; they generate massive amounts of customer data, given the 'always on' nature of customer usage, and provide a service that has become essential in many aspects of life. These advantages, coupled with what is perhaps the most significant technology change for the industry – 5G – can prove to be a force multiplier. While the telecom industry has undergone many technology upgrades, each more significant than the previous, 5G has everyone in the industry excited like never before. And there is a good reason for it.

Lesson 2: Why are Telecom operators obsessed with deployment of 5G?

- Capex investments made by telcos in 3G and 4G in the last decade have provided a major technological shift in the form of high-speed global mobile networks. We have already discussed how this has allowed the entry of OTT players who, leveraging the connectivity infrastructure of the telco operators, have grown revenues exponentially over the past decade, while telecom operators themselves haven't profited in any substantial way from this 5G, quite like its predecessors, is a huge leap forward in terms of speed, capacity and latency. But when combined with the distributed architecture and real-time Edge computing capability, this technology is unmatched. The Edge computing infrastructure allows telcos to shift the data processing and application deployment to devices they control, thus changing their equation with OTT providers.
- The connectivity needs of enterprises are rapidly evolving. Enterprises are no longer interested in static services; what they want is the ability to subscribe to varying speeds based on differentiated QoE (quality of experience), or the ability to use varying network capacity/slices, with the option to enhance or downsize network resources dynamically. The real power of 5G technology lies in its capability to enable and provide such transformational services for enterprises and industrial customers.
- However, for telcos to win in this scenario they will need to leapfrog in terms of product design to deliver world-class experience, including IOT devices and applications, to their customers. This will require innovative collaboration with OEM manufacturers, building a thriving ecosystem of tech startups to build pointed applications, and upgradation of their own product development and GTM infrastructure to stitch all these elements into compelling solutions for different industry-based enterprise customers and end consumers.

5G will mean a massive investment on the part of telecom players. The reason they are so excited about this is because it can potentially tilt the scales back in their favour on the customer's share of wallet. It opens up new possibilities in terms of new products and services that they can offer, thus driving new revenue streams. At the same time, 5G also provides them a way to drive legacy modernization across the OSS/BSS (operations support systems/business support systems) stack, which can drive cost optimization and unshackle their organizations from their decades-old systems and processes.

However, capitalizing on these advantages and the promise of 5G will also require telcos to focus on the following:

- Stronger push on digital channels
- Monetization of their massive data assets
- Legacy modernization to unshackle the organization
- Leveraging AI to build innovative solutions and enter new markets

Lesson 3: Digital channels

Digital channels can drive massive efficiencies and unlock huge dollars stuck in large physical assets, but this requires a different approach:

- *Products and services simplification* – Telcos have complex products, plans, pricing and processes – from legacy plans running into tens of thousands of lines and complex purchase journeys on the web to pricing plans that require an agent to decipher for the customer. Digital forces simplification, as customer adoption of unassisted channels dips if products are non-intuitive and puts a high cognitive load on customers. This simplification can also drive efficiencies overall by rationalizing the tech stack and the processes required to support the products.
- *Omni-channel strategy* – Telcos can convert their physical stores into enablers of digital sales through mechanisms like in-store/ kerb-side pick-up, local delivery (from store), etc. This will also require a redesign of incentives such that channels are not competing with but complementing each other. Disney, Target, Starbucks, Walgreens, etc., are good examples of traditional brick-and-mortar businesses that have successfully transformed into omni-channel experiences for the consumer and are fighting the e-commerce threat.
- *Customer Service* – This is another area where digital transformation can unlock immense value. Digital natives like Amazon, Uber, etc. drive over 90 per cent of their customer service through digital self-serve mechanisms and are constantly pushing the boundaries on use of AI/ML to predict and prevent customer service issues. Telcos, on the other hand, lie at the other end of the spectrum, with assisted customer service being a large cost head on their P&Ls.

Lesson 4: Data Assets

Telcos generate a huge amount of data; conversion of this data into powerful insights can allow them to build collaborative domain-specific solutions.

- Telcos, thanks to their unique positioning in the ICT (information and communications technology) value chain, have access to the deepest and richest insights into customer behaviour and usage. Telcos are also now able to serve these insights on a real-time and targeted basis using advanced AI/ML technologies for multiple use cases that extend beyond the telecom industry.
- To capitalize on this capability, telcos are developing external partnership models to leverage data for creating innovative products and services in digital ecosystems. Examples of this include:
 - Footfall analytics for more effective marketing
 - Credit scoring as a service
 - Health analytics using wearable technology, powering healthcare and insurance applications
- To fully unleash the potential of data, Telcos will need to build a more entrepreneurial culture and mechanisms. They will need to transcend their functional silos to build cross-functional PODs that are empowered to take product design and GTM decisions, and at the same time be accountable for the commercial success of these solutions.

Lesson 5: Modernization

Modernization of the legacy OSS (operational support system)/ BSS (business support system) infrastructure is a key enabler for telecom operators

Telecom operators are desperately looking to create new revenue streams, and with the advent of 5G, newer technical capabilities like network virtualization and network slicing will create possibilities for telcos to develop innovative digital connectivity solutions. However, unless the traditional OSS/BSS systems and applications are modernized to handle operations and billing at these newer revenue-generating services, telecom operators will not be able to obtain the desired revenue growth.

Traditionally, these OSS/BSS systems and applications have been operating in silos, with static and manual workflows. It is imperative for telcos to modernize the decades-old OSS legacy infrastructure. This will allow them to dynamically allocate network resources for their customers on a real-time basis. This capability will help them create new service offerings for existing and new customers. Along with OSS modernization, modernization of the billing stacks is also necessary so that these dynamically provisioned self-services, at a massive scale (connecting billions of devices), can also be billed to customers in accordance with the SLA commitments.

Most of the new service offerings from telecom operators to enterprise customers asks for planning, provisioning and operating in a real-time scenario. This implies the following requirements:

1. Modernization of network planning and optimization function.
2. Fundamental shifts in service provisioning and fulfilment applications for handling of real-time service offerings to customers.
3. Reimagination of network management and billing systems for operating across multiple network slices, both in physical and virtual domains.
4. Service assurance and customer experience management systems to be readied for data flow in both directions, i.e., from network to users and from users back into the network.

Lesson 6: Artificial intelligence (AI)

AI can drive efficiencies, improve customer experience and be a source of product and service innovation.

AI is changing the way the world does business, and it is no different in the telecom industry. The key areas where telcos are investing in AI include:

- *Improvement of customer service through chatbots and virtual assistants*
Telcos get a massive number of customer service requests for installation, troubleshooting and maintenance. Virtual assistants automate and scale responses to these requests at high speeds, which dramatically cuts business expenses and improves customer satisfaction. Vodafone's chatbot TOBi and Nokia's virtual assistant MIKA or Telefonica's Aura are good examples of telcos leveraging chatbots and VAs to deliver superior customer experience at lower costs.

- *Network quality improvement using predictive maintenance and self-organizing networks (SON)*
Prevention of outages using AI/ML predictive models can enhance customer experience while delivering huge cost savings by preventing costly support calls and reactive truck rolls.
 - *Fraud Detection and Prevention*
Fraud is a big driver of costs and bad customer experience. To prevent fraud, telcos end up putting restrictive policies, checks and balances that make life difficult even for their genuine customers. AI/ML allows companies to leapfrog in their ability to predict and detect fraud using the sophisticated models and massive data at their disposal. They can now implement more nuanced controls that are razor-focused on fraudulent activity.
 - *Personalized recommendations*
As telcos shift more of their business to digital channels, AI- based predictive models allow them to build more personalized offers and purchase journeys, thereby improving overall adoption of digital channels and improving KPIs like revenue, cross- selling/upselling, etc.
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Nitin's conclusion:

5G is certainly a game-changer for telecom operators because it has the ability to open new revenue fronts, at the same time also shifting the balance of customer ownership in their favour from the OTT players. However, in order to capitalize on this opportunity, telcos will have to ensure timely modernization of legacy Infrastructure, develop digital channels more effectively and adopt emerging technologies like AI and ML to deliver world-class customer experience.